Ordinary Differential Equation (ODE) Solvers

James Rosado[1]

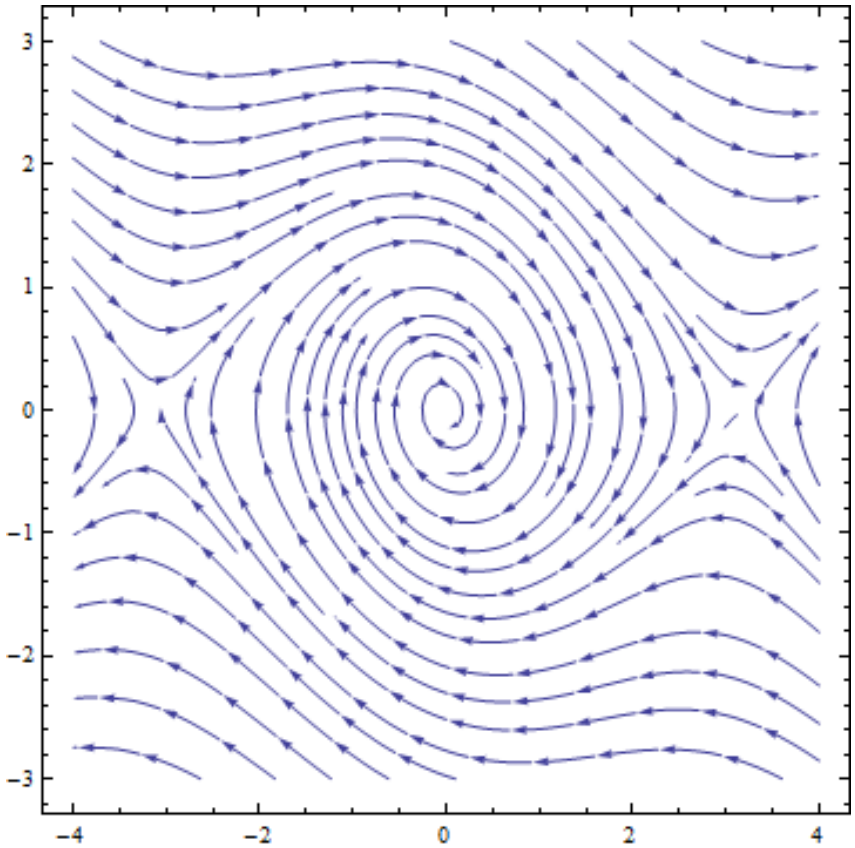
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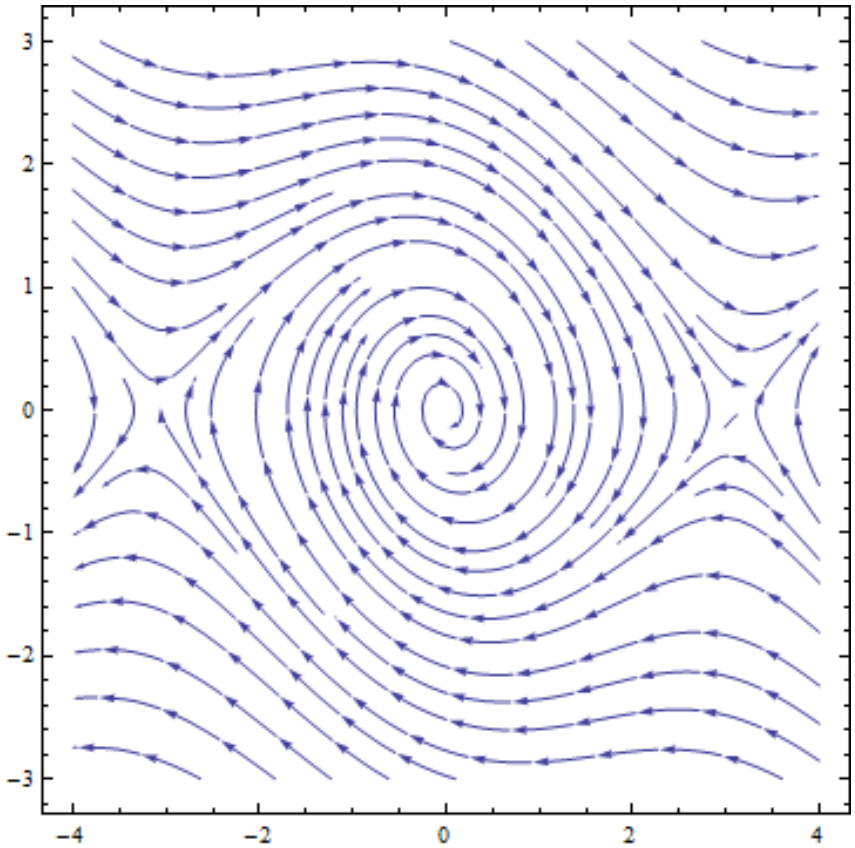
Outline

1. ODE Basics
   - What is an ODE?
   - Example ODE
   - Types of ODEs
   - Stability and Stiffness
2. ODE Solvers
   - What is a solver?
   - An example
   - Which solver and when?



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What? & Why?

A general n-th order ODE with constant coefficients is of the form:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{dy^{n-1}}{dx^{n-1}} + \cdots + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} = f(x, y), \qquad (1)$$

and will have n initial conditions

$$y(x_0) = y_0, y'(x_0) = y_1, \cdots, y^{(n)}(x_0) = y_n.$$

We consider *first-order* ODEs [4] of the form

$$y' = \frac{dy}{dx} = f(x, y) \text{ with } y(x_0) = y_0, \qquad (2)$$

where y is a vector of dimension d.

What? & Why?

Since $y = \{y_1(x), y_2(x), \ldots, y_d(x)\}$ is a vector then we have a set of first-order equations

$$y' = \begin{pmatrix} y_1'(x) \\ y_2'(x) \\ \vdots \\ y_d'(x) \end{pmatrix} = \begin{pmatrix} f_1(x, y_1, y_2, \ldots, y_d) \\ f_2(x, y_1, y_2, \ldots, y_d) \\ \vdots \\ f_d(x, y_1, y_2, \ldots, y_d) \end{pmatrix}.$$

High order ODE problems like (1) can always be reduced to a set of first-order ODEs.

What? & Why?

- Arise in physics, astronomy to model celestial mechanics and motions of particles.
- Used in chemistry to model reaction rates.
- In biology ODEs are needed to determine the rate of spread for various infectious diseases.
- ODEs are also found in economics to measure interest rates and predict stock trends.
- Integrate & Fire model, Hodgkin-Huxley model,...for describing the action potential across a neuron membrane.

ODEs model many natural phenomena to predict changes, but it is very difficult to solve (if not impossible) these ODEs analytically.

Predator-Prey

Let $y_1(t) =$ population of predator at time t and $y_2(t) =$ population of prey at time t.

Population growth rates are related (Lotka-Volterra [7]) as follows

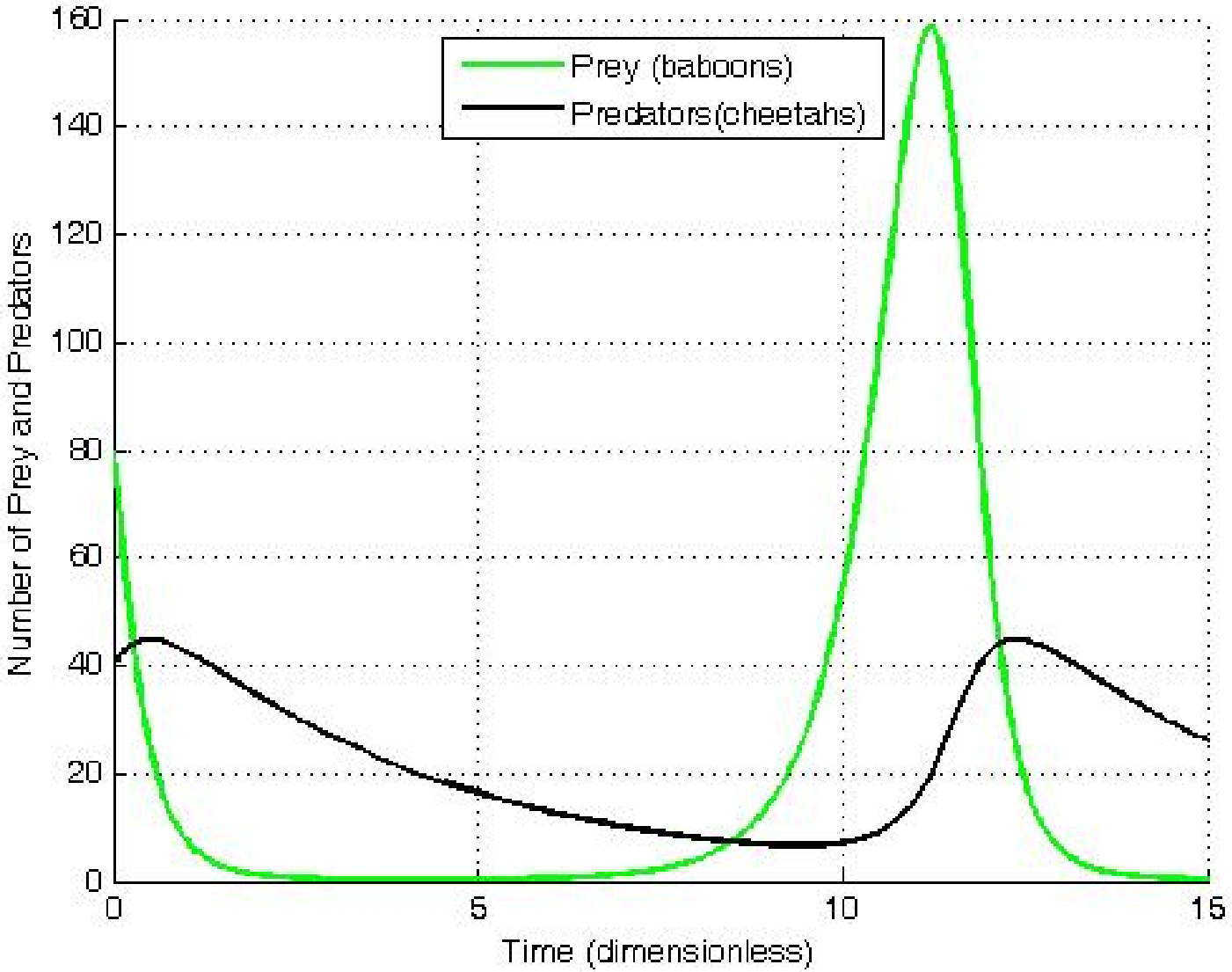
$$y_1'(t) = \alpha y_1 + \beta y_1 y_2,$$

$$y_2'(t) = \delta y_1 y_2 - \gamma y_2;$$

with initial populations $P_1 = y_1(t_0)$ and $P_2 = y_2(t_0)$.
This set of equations can be written in the form (2)

$$y'(t) = \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} f_1(t, y_1, y_2) \\ f_2(t, y_1, y_2) \end{pmatrix} = f(t, y).$$

Predator-Prey



Neuronal Cell Membrane Potential

Hodgkin-Huxley [6] model:

$$C_m \frac{dV_m}{dt} = \sum_j \mathrm{I}_j(t, V_m),$$

where $V_m(t) =$ the membrane potential and $\mathrm{I}_j(t, V_m) =$ is the current induced by the flow of ions traversing the membrane.

Each ion induces a current:

$$\mathrm{I}_{\mathbf{Na}^+}(t, V_m) = g_{\mathbf{Na}^+}(V_m(t) - V_{eq}),$$

$$\mathrm{I}_{\mathbf{K}^+}(t, V_m) = g_{\mathbf{K}^+}(V_m(t) - V_{eq}),$$

and $g_{\mathbf{X}} =$ conductance for particular ion, $V_{eq} =$ resting potential of the membrance.

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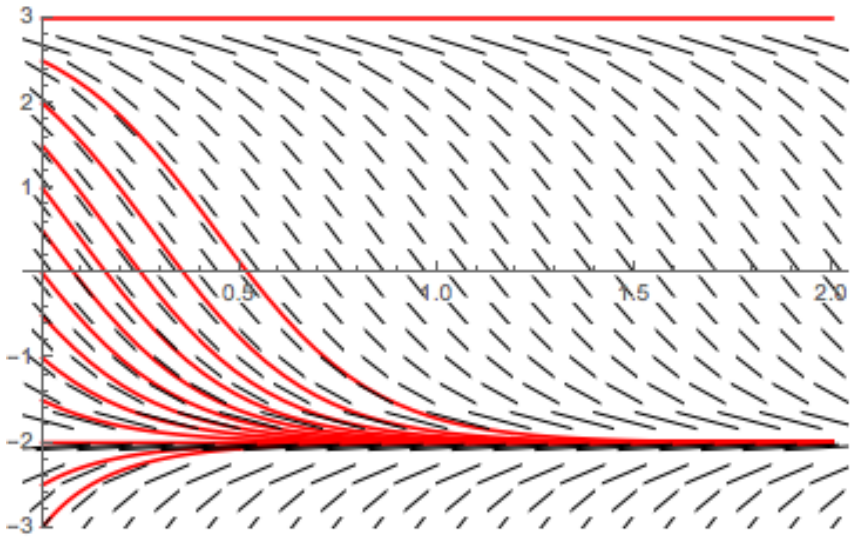
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- **Stiff** ODEs are numerically *unstable* for certain numerical algorithms...
- **Non-stiff** ODEs have numerical solutions that are easily obtained using *classical* methods.

Stability

- An ODE has what are called *equilibrium* points.

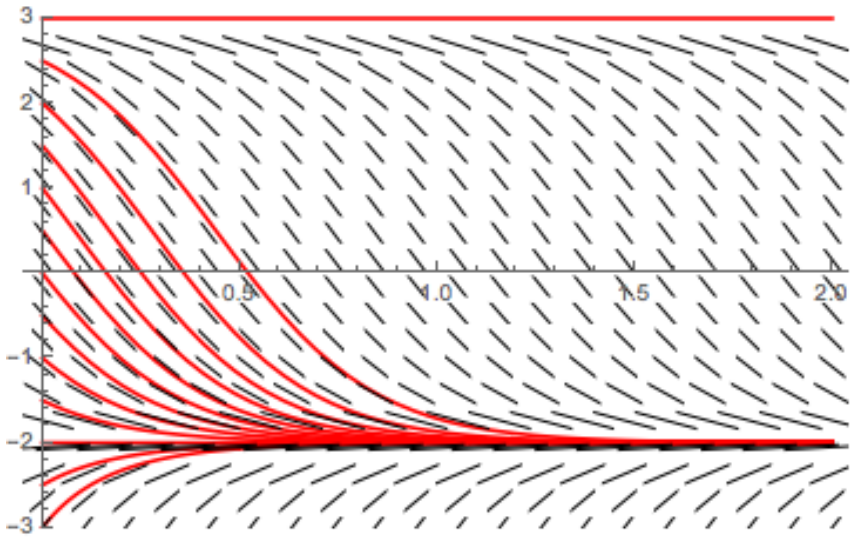
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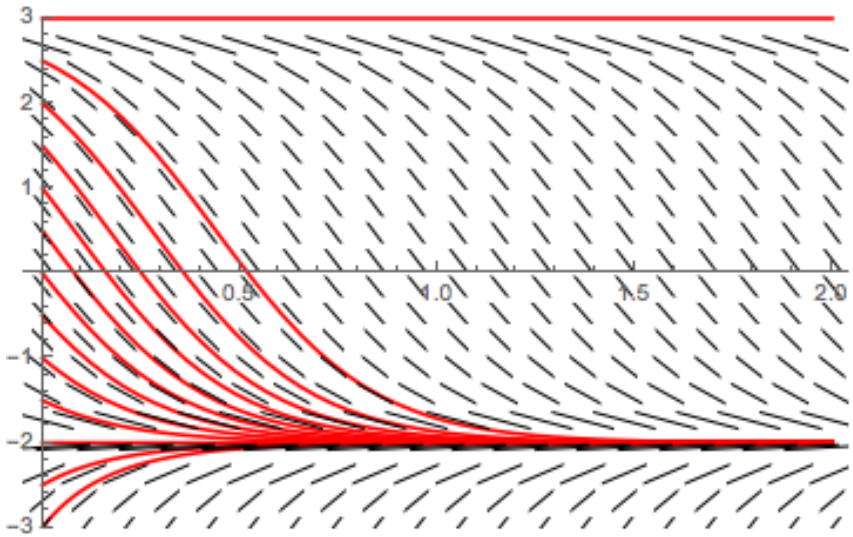
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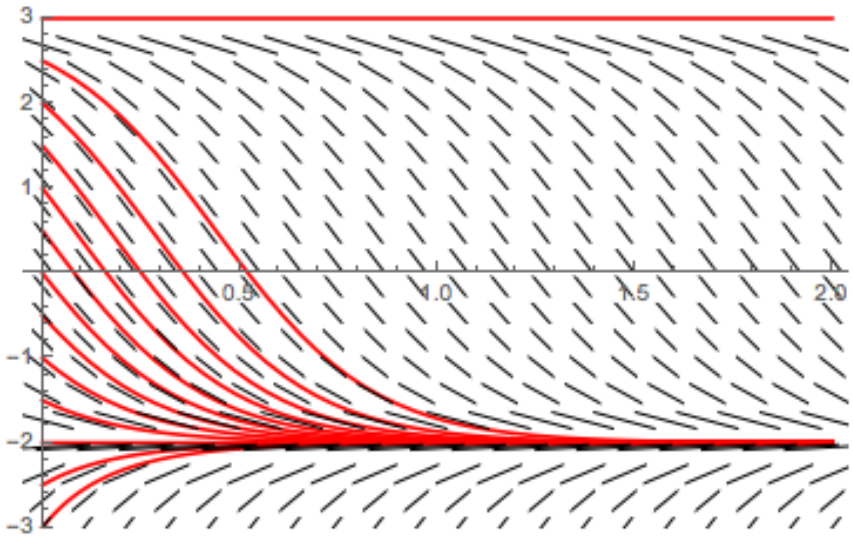
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- ODE is unstable if small moves away from equilibrium points yields significant changes in the rate of change.

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- If $y' = f(x_0, y_0) = 0$, then (x_0, y_0) is an equilibrium point.
- Re-write the system as a linearization near (x_0, y_0):

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} f_1(x, y_1, y_2) \\ f_2(x, y_1, y_2) \end{pmatrix} \approx \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \mathbf{J}_0 \begin{pmatrix} u_1 \\ u_2 \end{pmatrix},$$

where $\mathbf{J}_0$ is the Jacobian [1]: $\begin{pmatrix} \frac{\partial f_1}{\partial x}(x_0, y_0) & \frac{\partial f_1}{\partial y}(x_0, y_0) \\ \frac{\partial f_2}{\partial x}(x_0, y_0) & \frac{\partial f_2}{\partial y}(x_0, y_0) \end{pmatrix}$.

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- If all spectral values of $\mathbf{J}_0$ have negative real part, then stable near equilibrium point [8].
- If one of spectral value has positive real part, then equilibrium point is unstable [8].

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Definition (Curtiss & Hirschfelder [3])

Initial value problems are stiff if they are (exceedingly) difficult to solve by ordinary, explicit step-by-step methods, whereas certain implicit methods perform quite well.

ODE Stiffness

- Rapidly changing slopes positive/negative away from stable solution.
- $\Delta x =$ *resolution*, step size, mesh size for numerical approx.
- From (2) let $f(x, y) = \frac{y - G(x)}{a(x,y)}$

$$\left| \frac{a(x, y)}{\Delta x} \right| \ll 1 \Rightarrow \text{ stiff.}$$

- Δx too big leads to large variation in slopes [3].

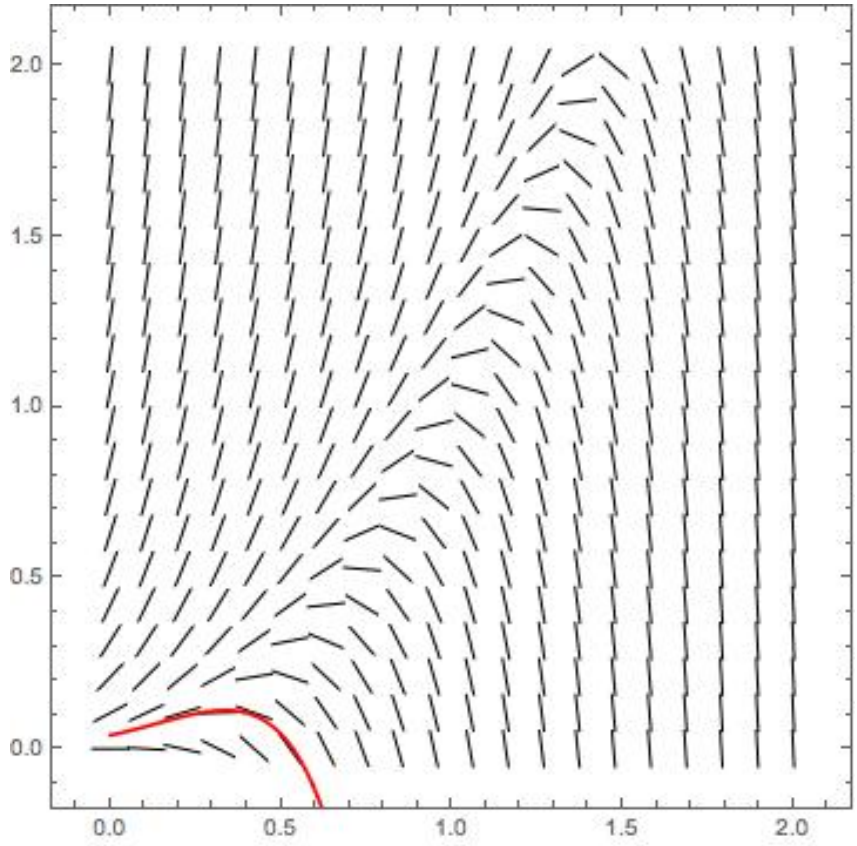


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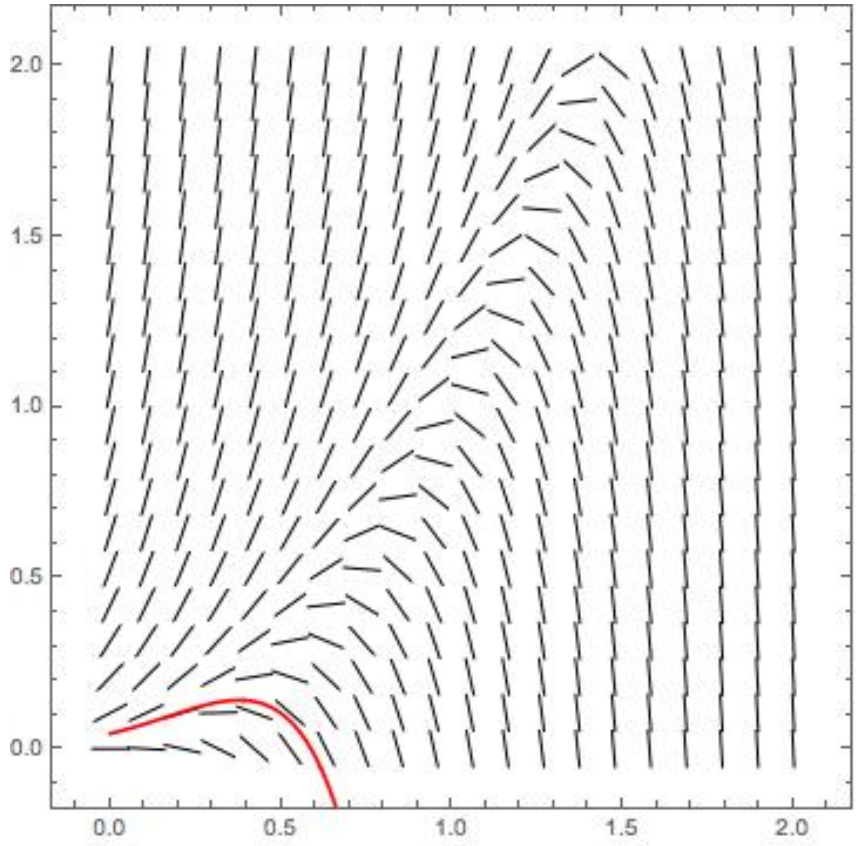


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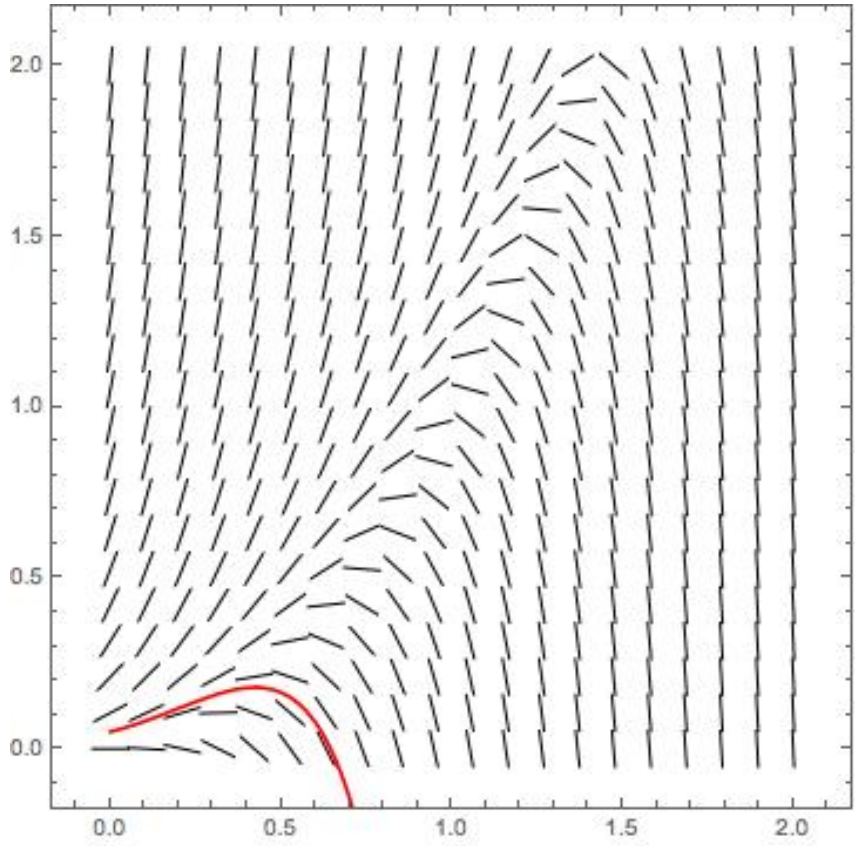


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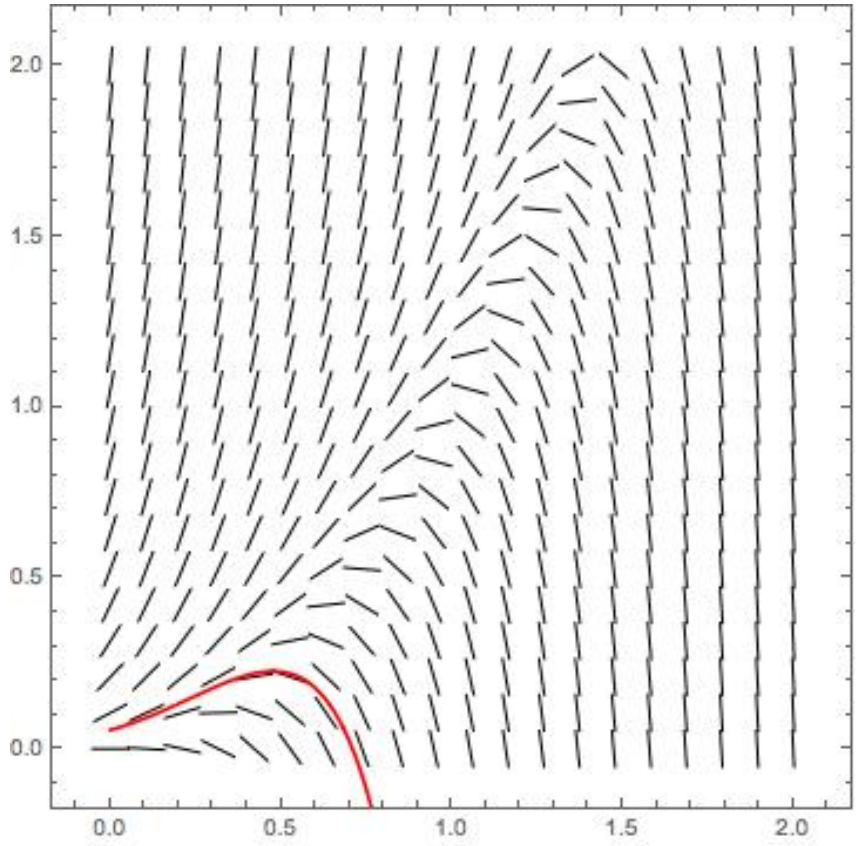


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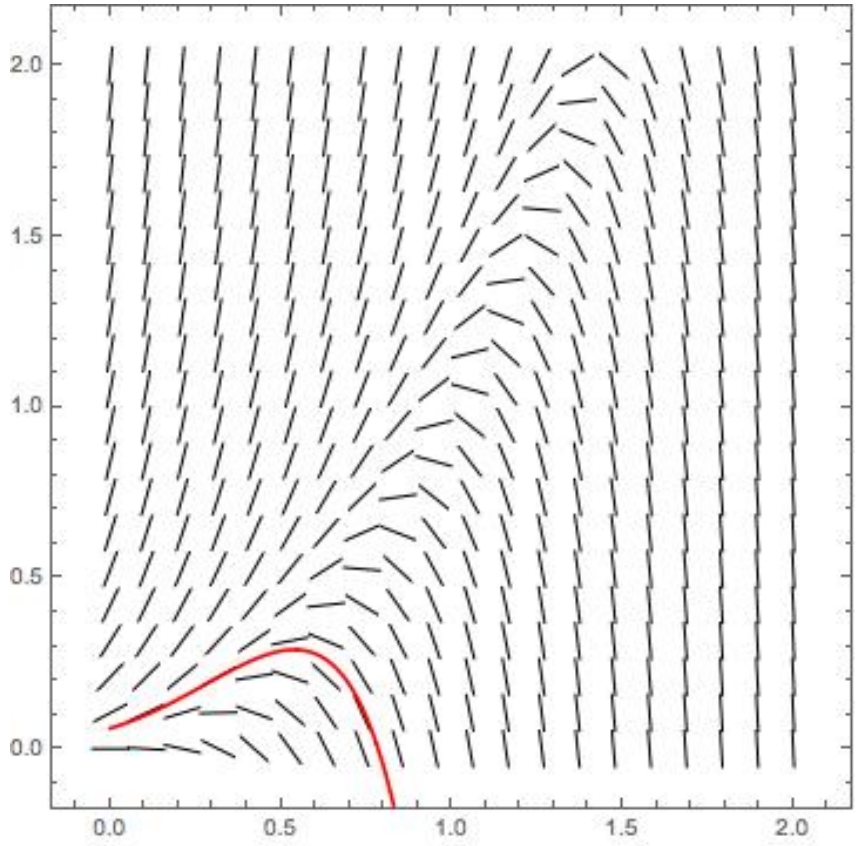


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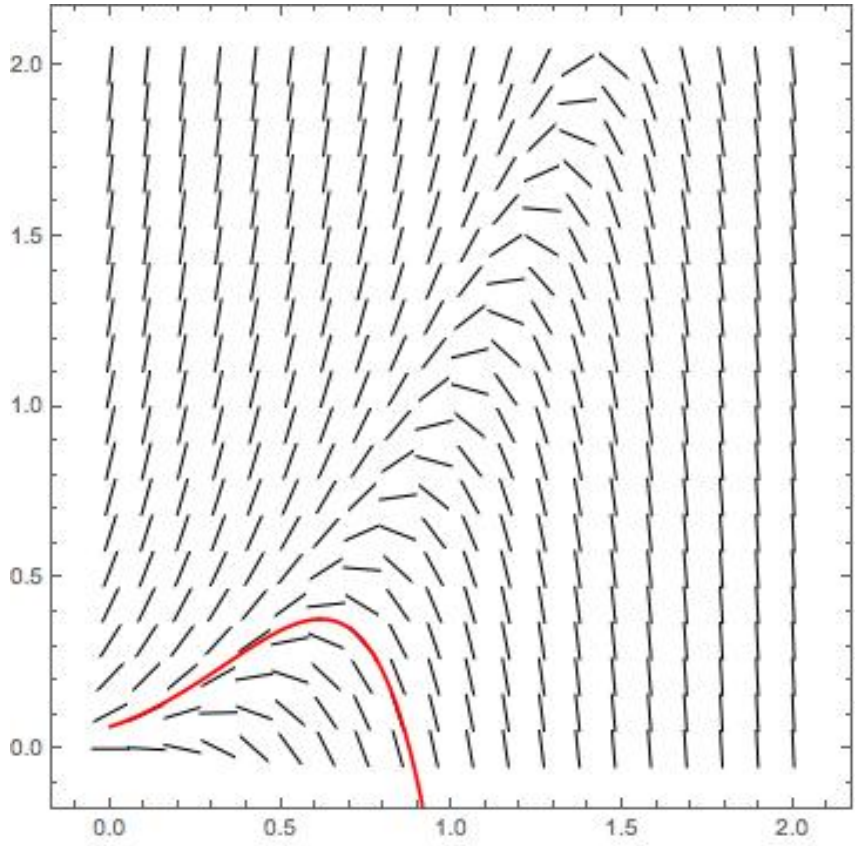


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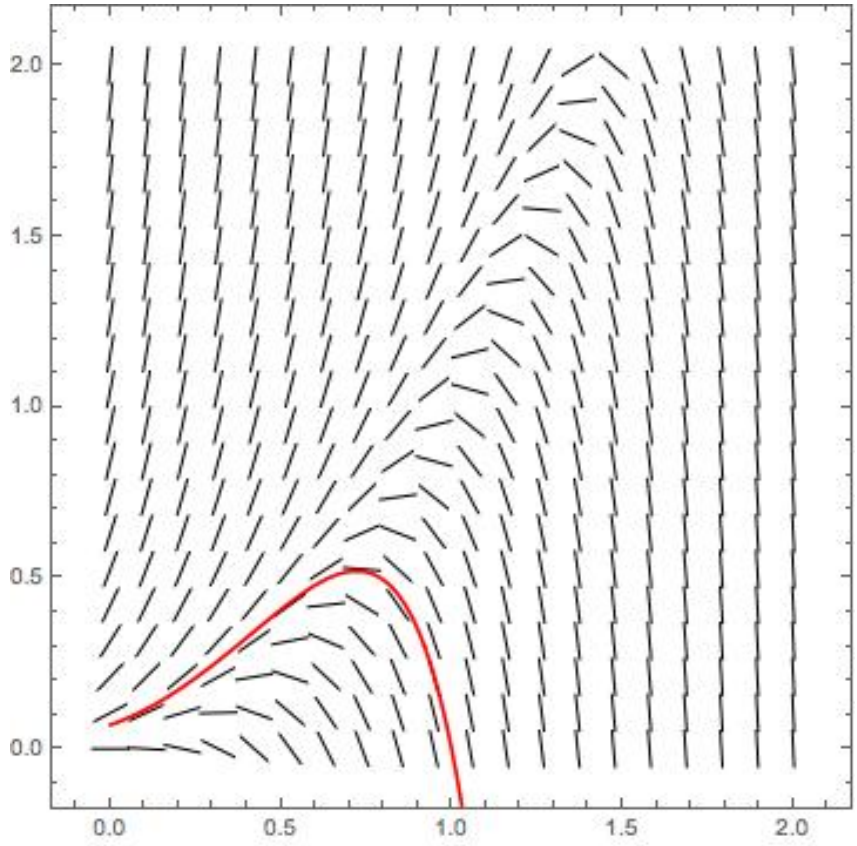


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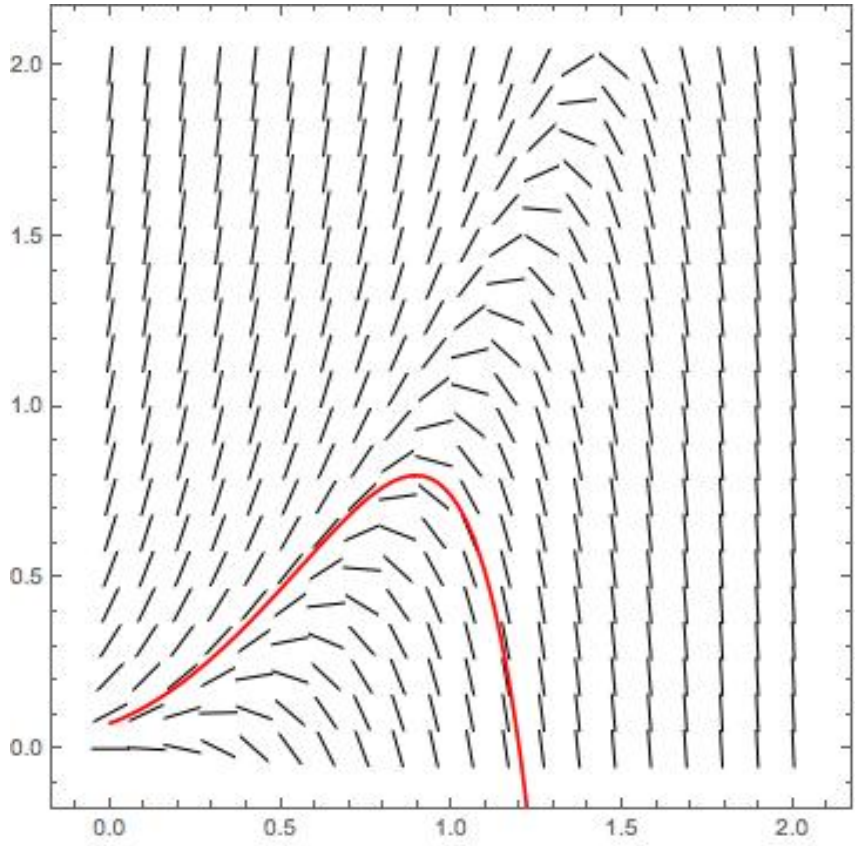


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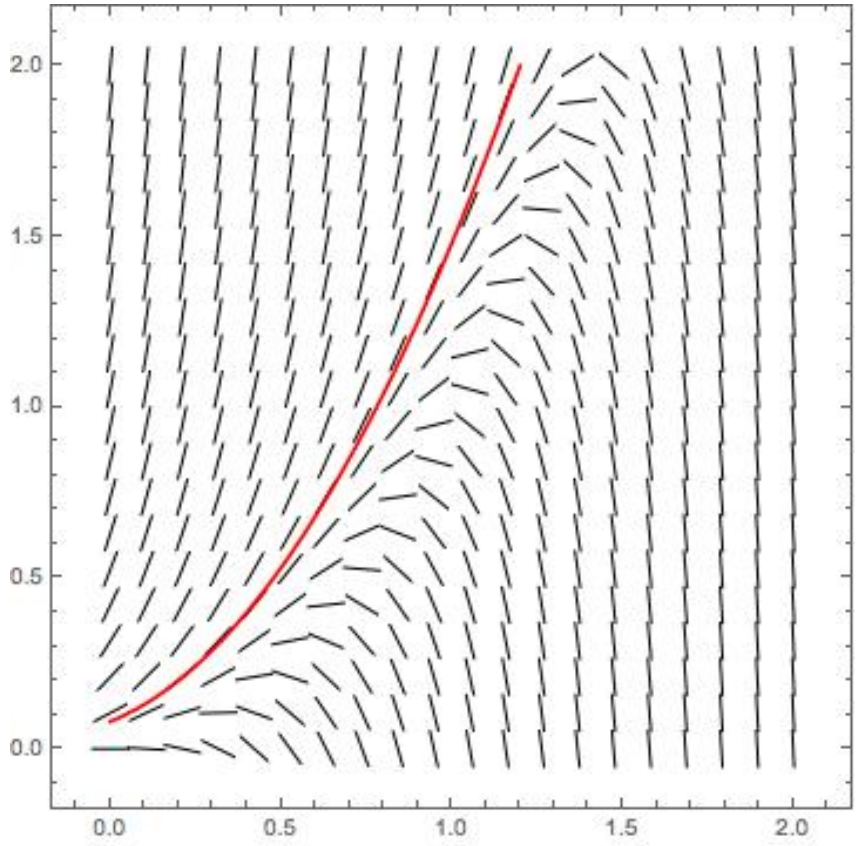


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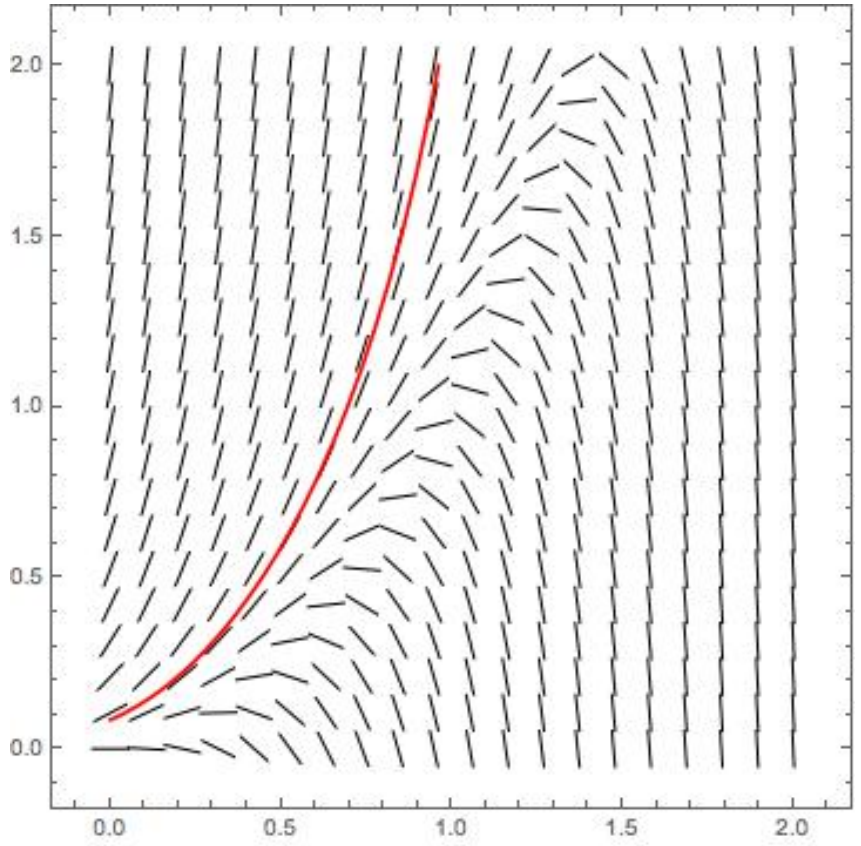


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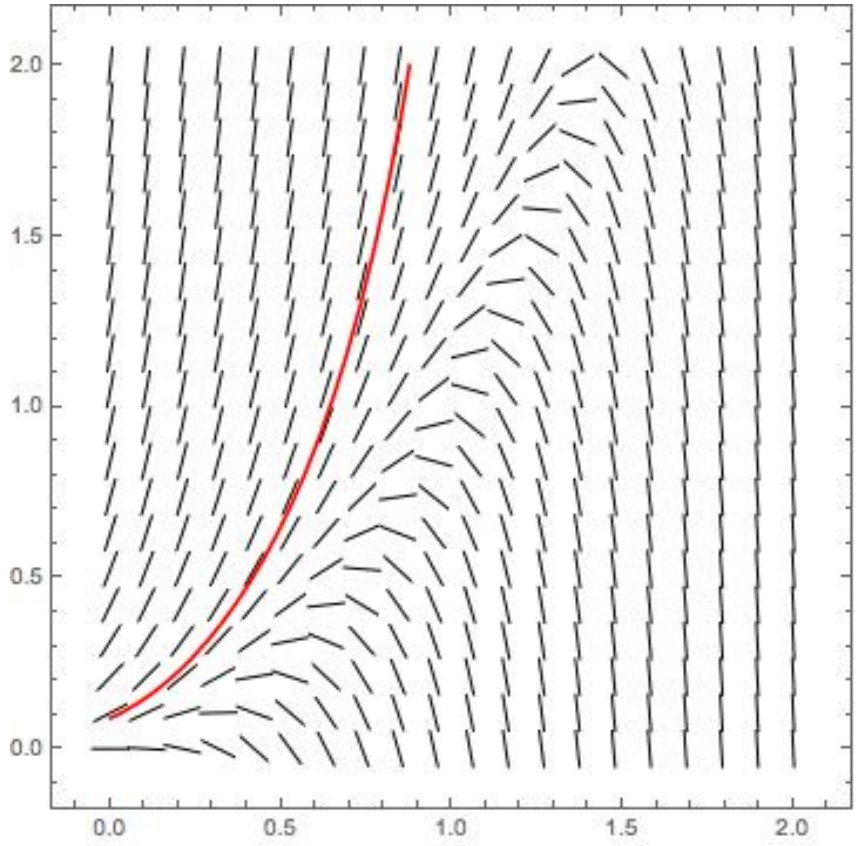


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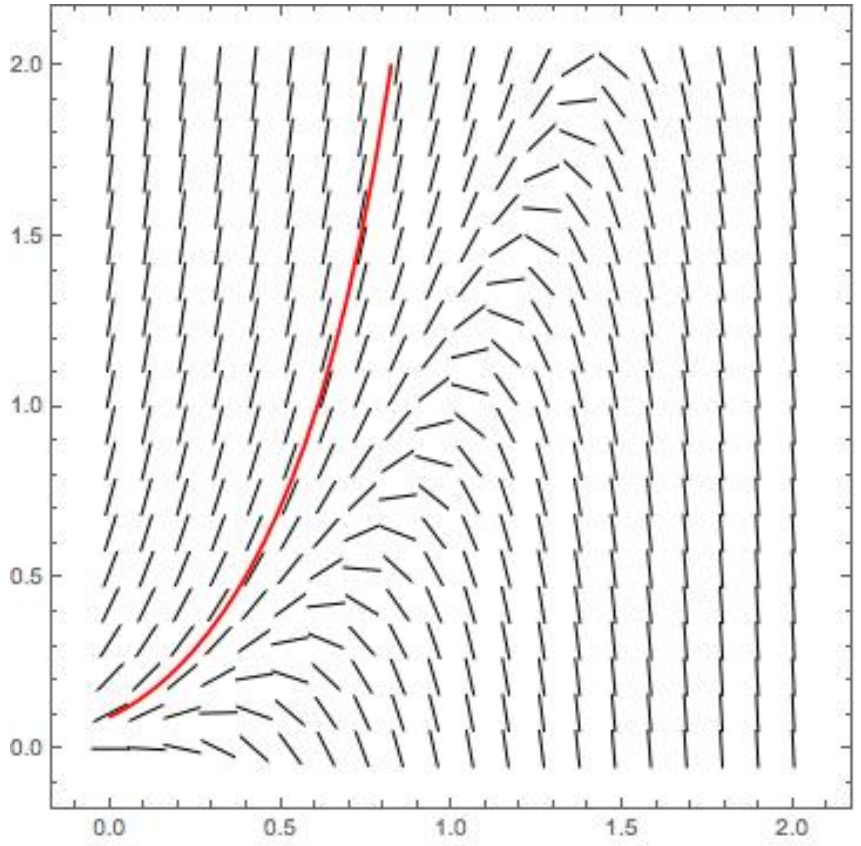


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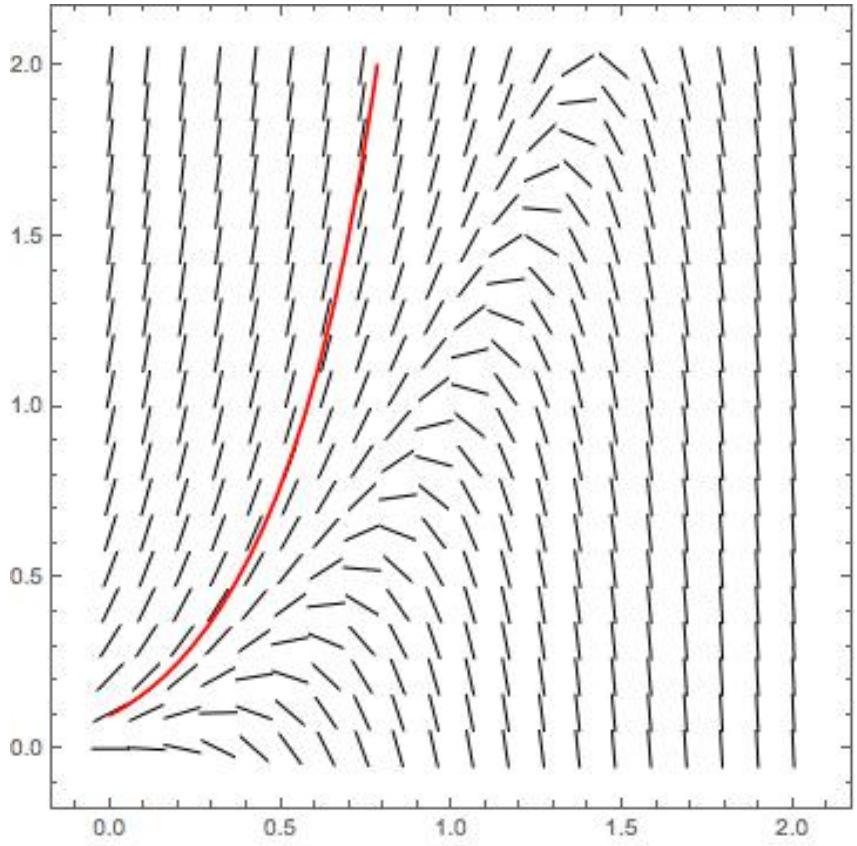


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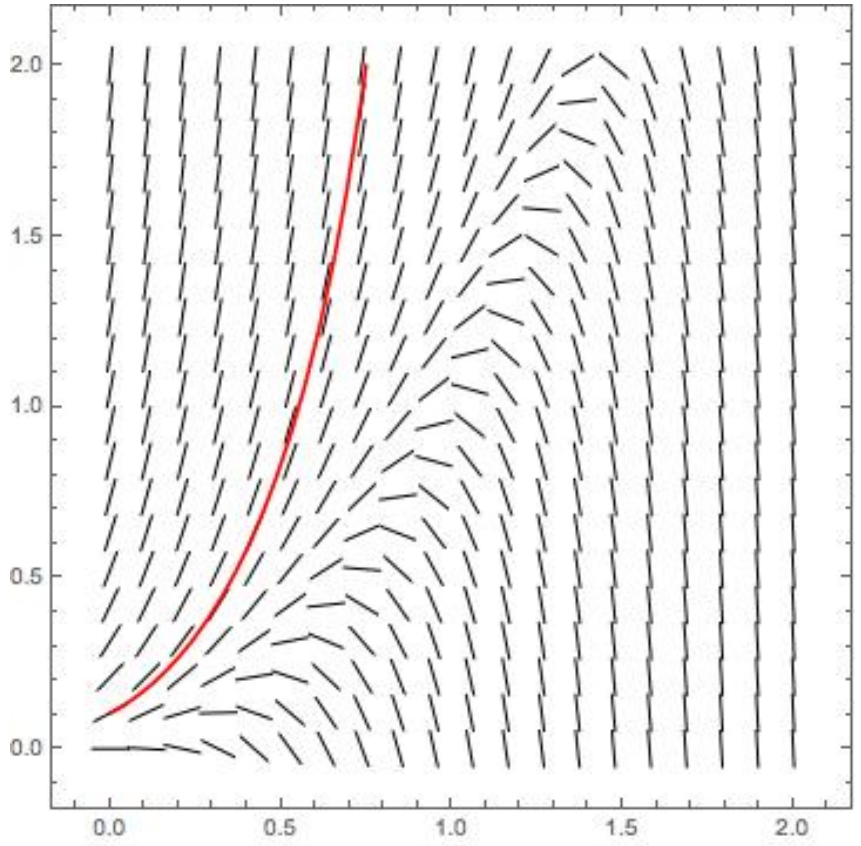


Figure: A stiff ODE $\frac{dy}{dx} = 5(y - x^2)$ from [3].

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- Rapidly changing slopes positive/negative away from stable solution.
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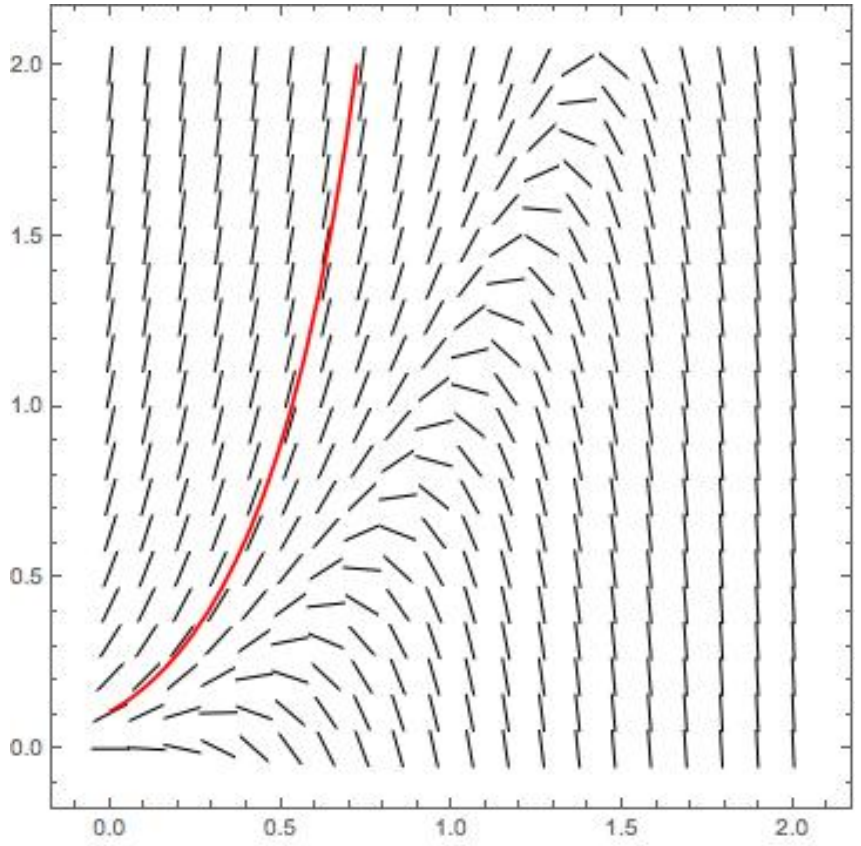


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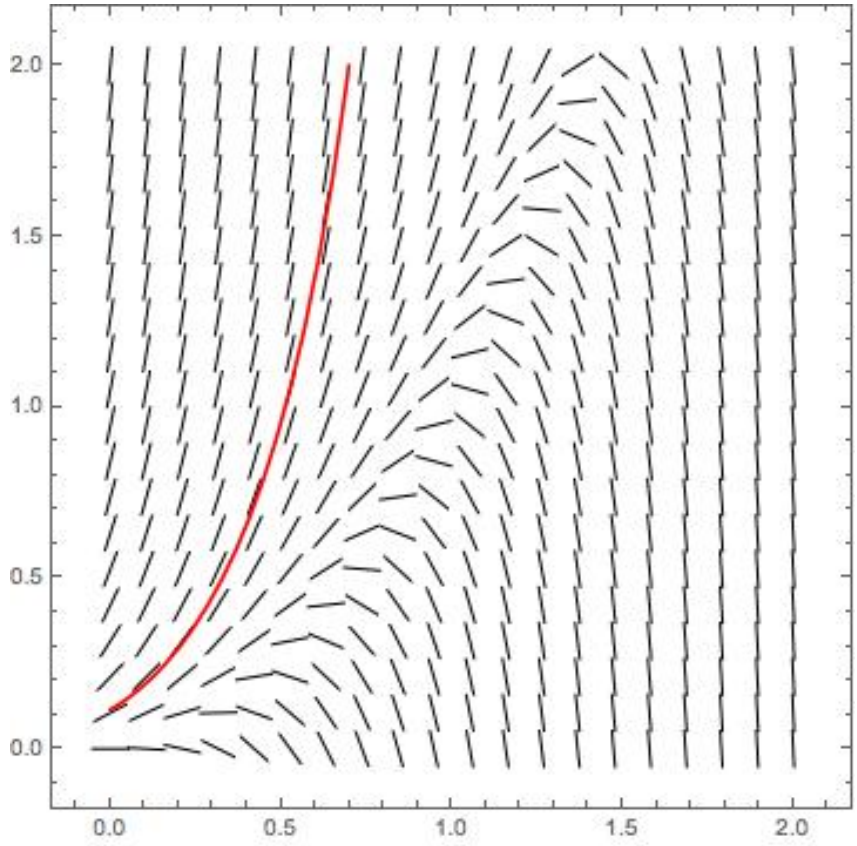


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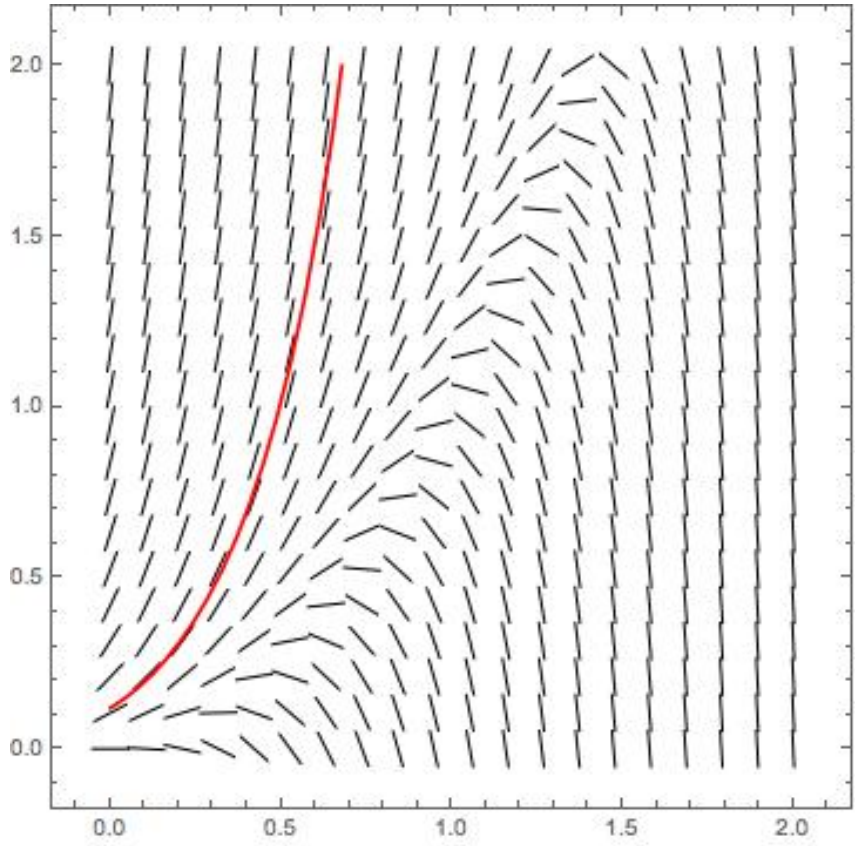


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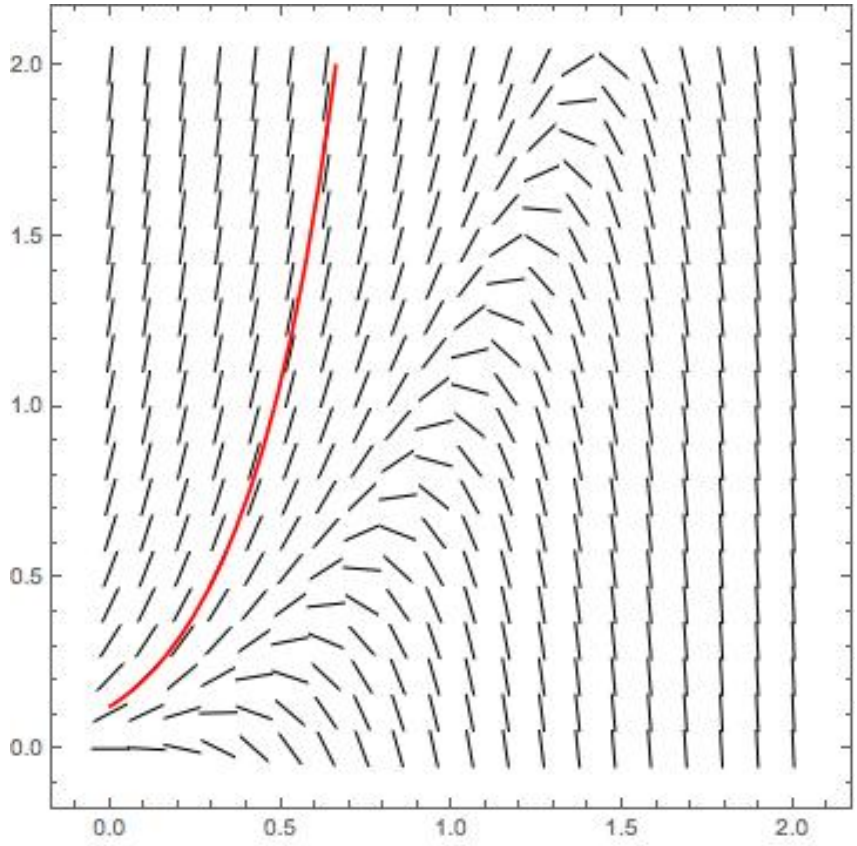


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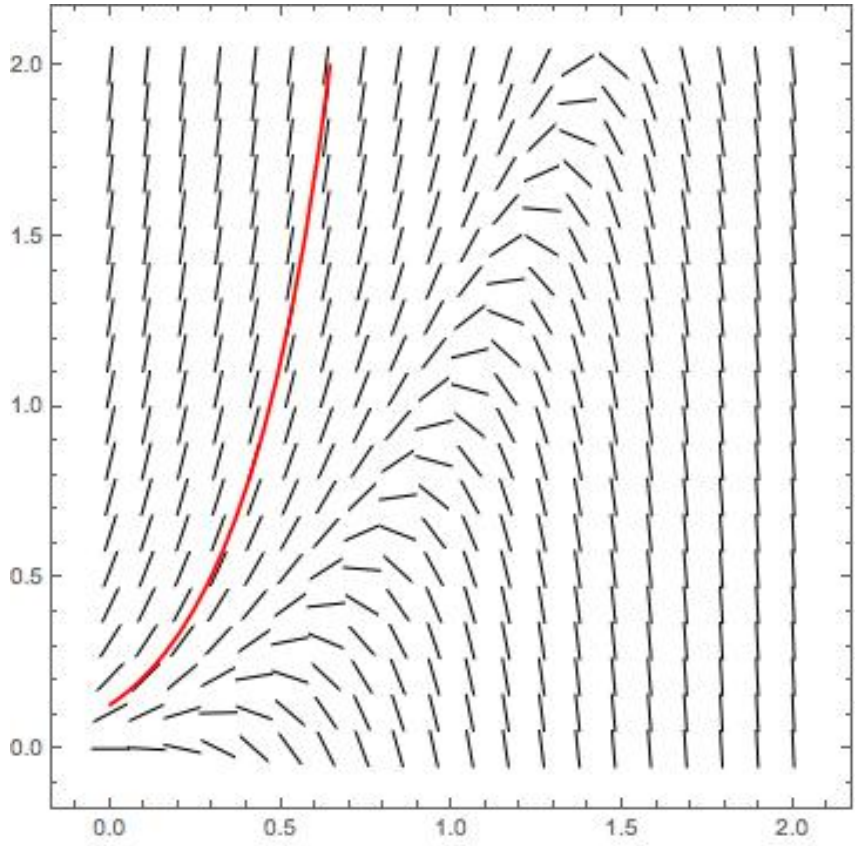


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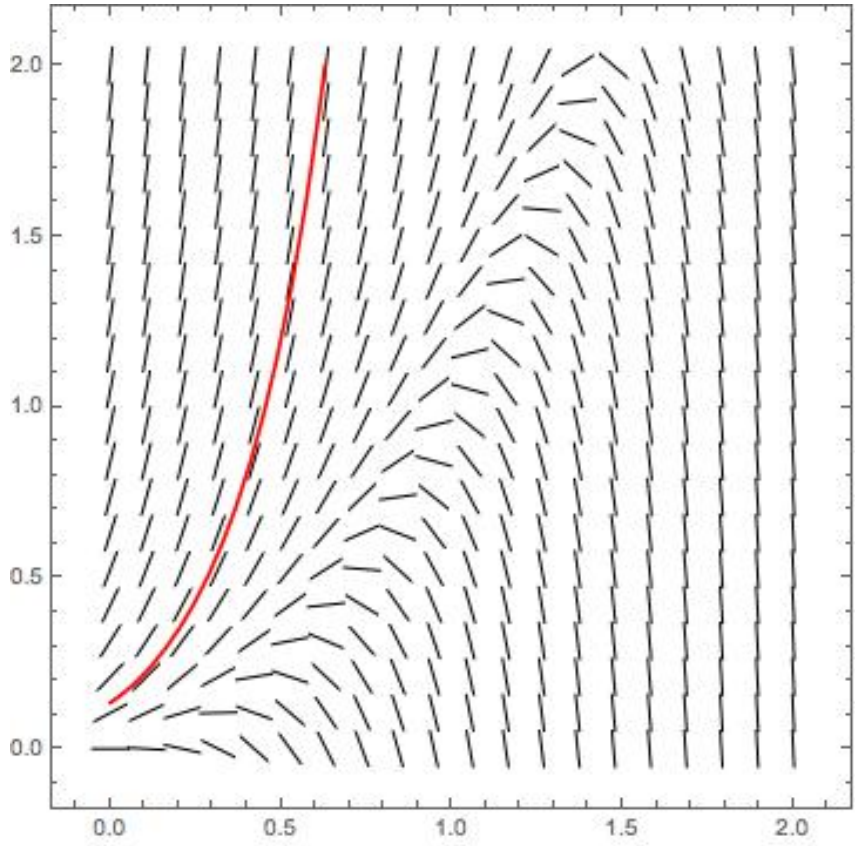
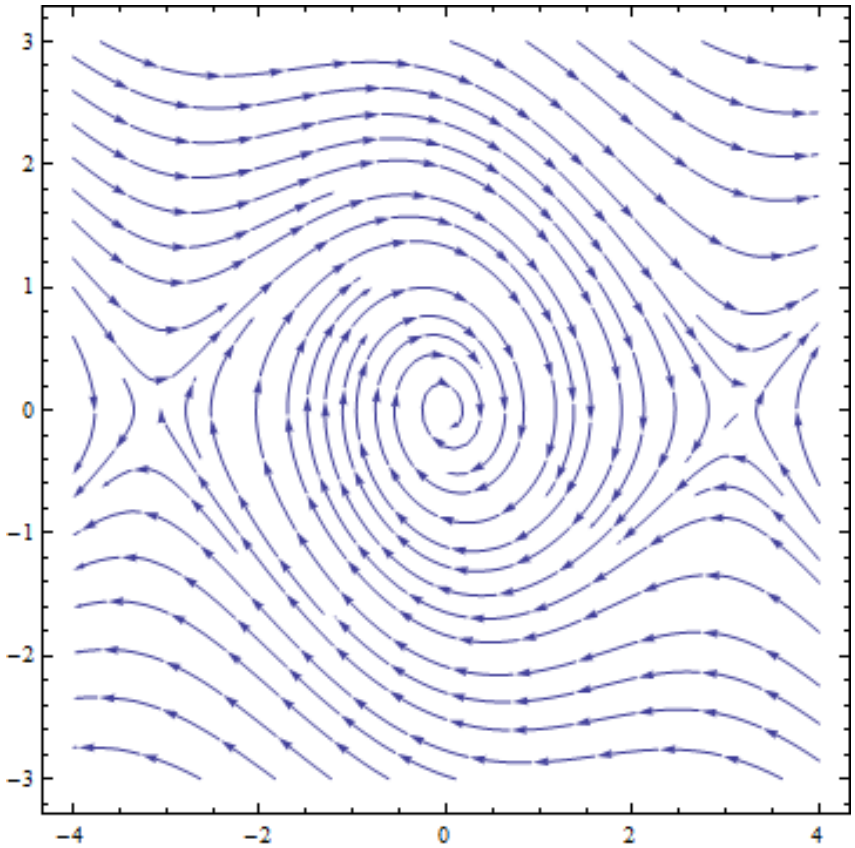


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Outline

1. ODE Basics
   - What is an ODE?
   - Example ODE
   - Types of ODEs
   - Stability and Stiffness
2. ODE Solvers
   - What is a solver?
   - An example
   - Which solver and when?



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- Problem: each iteration has a "cost", the number of computations!
- Smaller step size = better resolution, but more computations per iteration = more time.

Common ODE Solvers

Stiff ODEs

- Implicit Runge-Kutta Methods (RKF)
- Linear Multistep Methods (LMF)
- Backward Differentiation Formulas (BDF)
- Extrapolation Methods, trapezoidal rule
- Second Derivative Multistep Formulas
- Blended Linear Multistep Methods

Non-Stiff ODEs

- Classical RKFs: Euler, Backward Euler
- Runge-Kutta-Fehlberg
- LMF: Adam's formulas
- Richardson Extrapolation Method
- Taylor Series
- Multi-rate Methods

The ODE: $y'(x) = -15y(x)$ with $y(0) = 1$

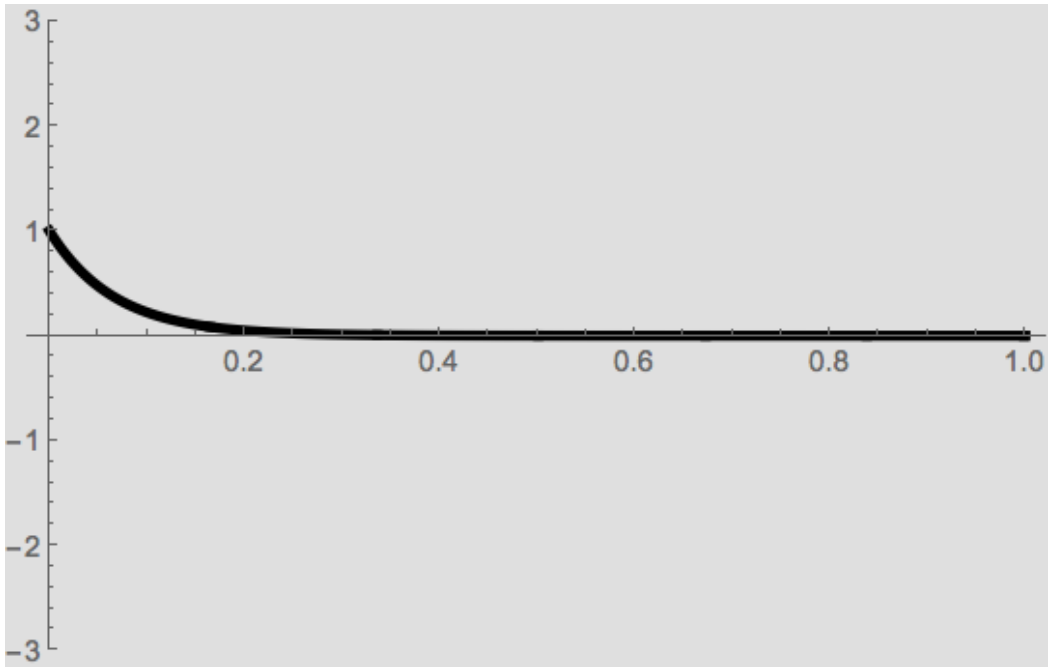


Figure: Solution curve: $y(x) = e^{-15x}$

$$\frac{dy}{dx} = -15y \Rightarrow \frac{dy}{y} = -15\ dx \Rightarrow \int \frac{dy}{y} = \int -15\ dx \Rightarrow \ln(y) = -15x + C$$
$$\Rightarrow y(x) = y_0 e^{-15x}$$

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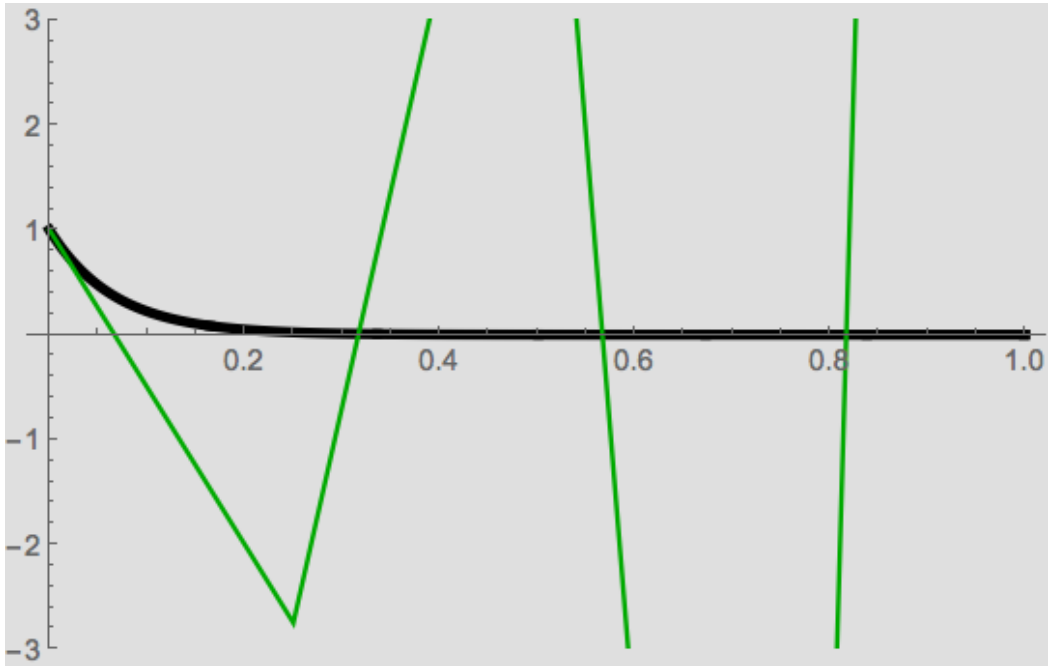


Figure: Euler with $\Delta x = 0.25$

With step size $\Delta x = 0.25$ over $[0, 1]$ we have points $\{0, 0.25, 0.5, 0.75, 1\}$.

$$y_{n+1} = y_n + \Delta x \cdot f(x_n, y_n) = y_n + \Delta x \cdot (-15y_n)$$

There is extreme variation among approximations.

The ODE: $y'(x) = -15y(x)$ with $y(0) = 1$

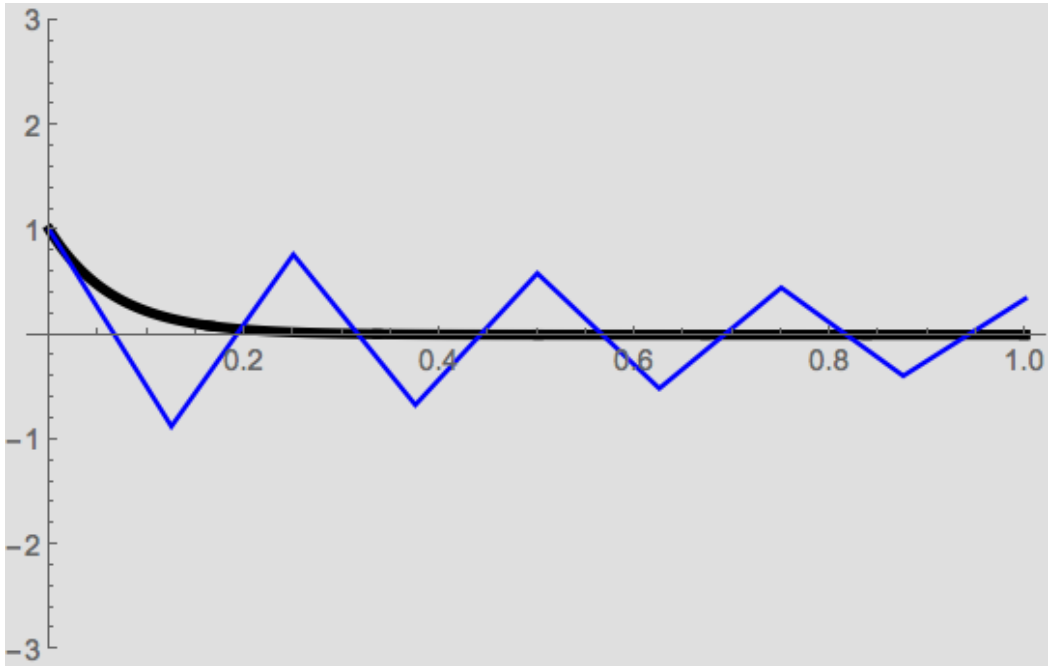


Figure: Euler with $\Delta x = 0.125$

With step size $\Delta x = 0.125$ over $[0, 1]$ we have points $\{0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1\}$.

We have a closer approximation, with less variation, because of smaller (half) step size = twice as many iterations.

The ODE: $y'(x) = -15y(x)$ with $y(0) = 1$

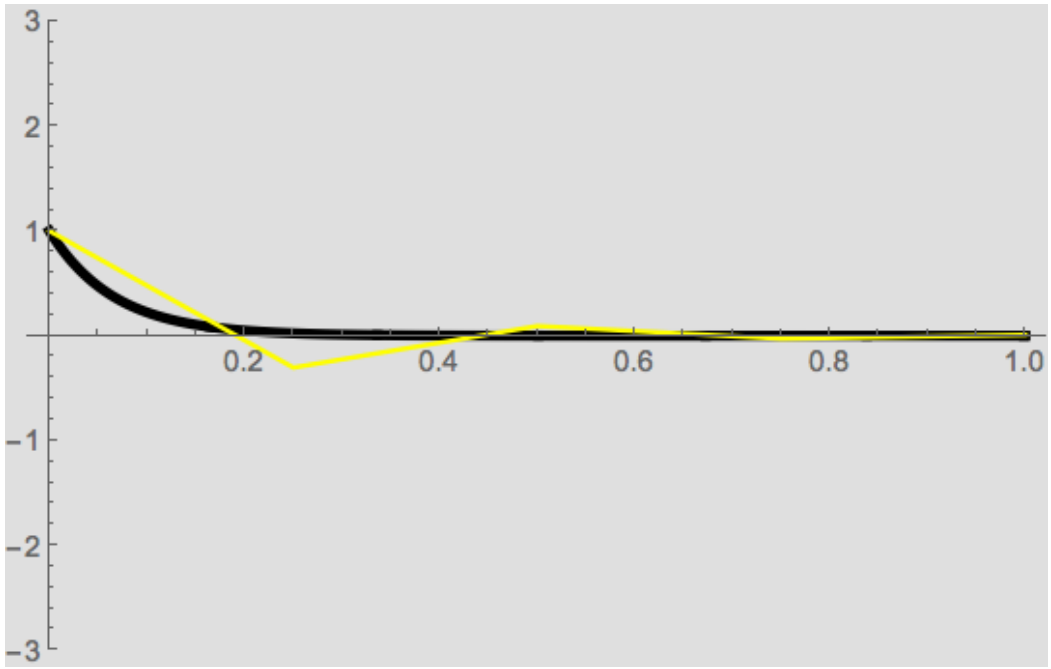


Figure: Adam's Method

Here, we let $\Delta x = 0.25$, with points $\{0, 0.25, 0.5, 0.75, 1\}$:

$$y_{n+1} = y_n + \frac{1}{2}\Delta x \cdot \left(f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right).$$

We achieve an even closer approximation with only 4 iterations, instead of 8 with Euler.

The ODE: $y'(x) = -15y(x)$ with $y(0) = 1$

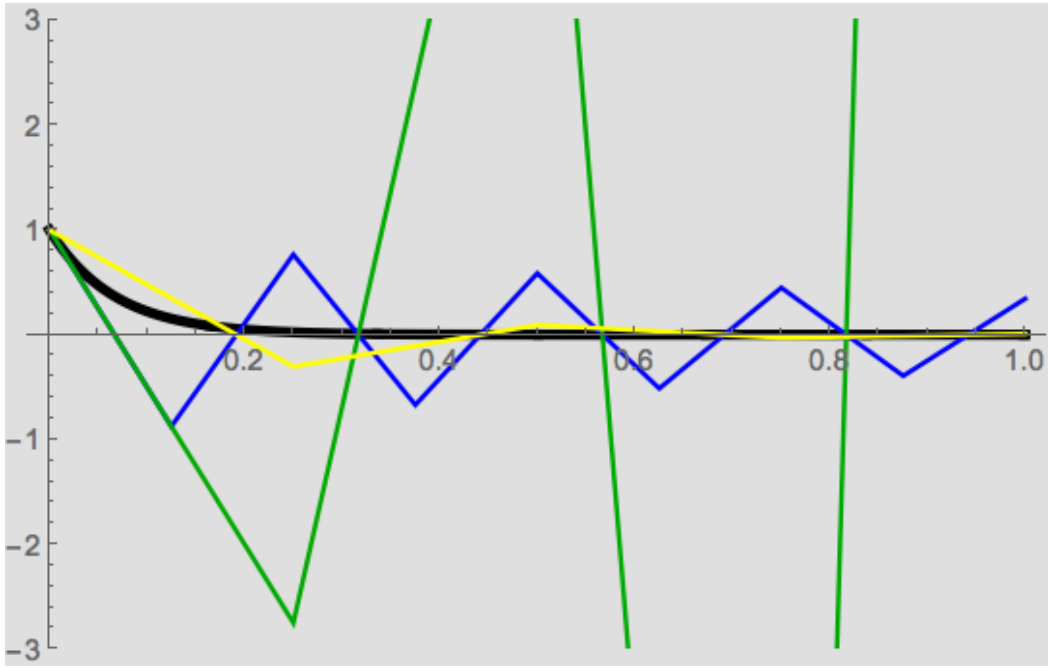


Figure: Full Comparison

This example was an example of a stiff ODE.

Notice for Euler method:

$$y_{n+1} = y_n(1 - 15\Delta x) = y_{n-1}(1 - 15\Delta x)^2 = \cdots = (1 - 15\Delta x)^k y_0,$$

where $(1 - 15\Delta x)^k$ grows geometrically if Δx is too large.

Best solvers to use.

Runge-Kutta Formulas (RKFs): Euler, Backward Euler...

- Explicit ODEs, no implicitly defined equations.
- Pair up RKFs (known as RK-Fahlberg).
- Cost of iteration $< 5\sqrt{\text{no. computations in equation}}$.

Linear Multistep Formulas (LMFs)

- Based on Adam's formula, like trapezoid rule and Simpson's rule.
- Well suited for applications with many output points.
- Most efficient when cost of iteration $> 50\sqrt{\text{no. computations in equation}}$
- Closer approximations in fewer iterations.

Implicit Methods

Definition (Adams-Moulton Four-Step [2])

Given: y_0, y_1, y_2, y_3 then

$$y_{n+1} = y_n + \frac{\Delta x}{720}\left(f(x_{n+1}, y_{n+1}) + 646f(x_n, y_n) - 264f(x_{n-1}, y_{n-1}) + 106f(x_{n-2}, y_{n-2}) - 19f(x_{x-3}, y_{y-3})\right).$$

Yields better approximations for implicitly defined ODEs, but it may not be possible to explicitly obtain y_{n+1} at each iteration.

With any method there are *local* and *global* errors that occur.

Local Errors: truncation error after one iteration of the algorithm.

Global Error: truncation error after a complete run of algorithm.

Summary

Considerations for choosing a solver...

- Number of calculations per iteration, degree of ODE.
- Type of stability of the ODE.
- Stiff vs. Non-stiff.
- Step size Δx, resolution.
- Explicitly or Implicitly defined ODEs.
- We have yet to consider ODEs with *non-constant* coefficients.

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